

Algorithms Portfolio

Curated algorithmic solutions with correctness and complexity analysis

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C++ primary • Python secondary

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1 Introduction

This document is a curated portfolio of algorithmic solutions. Each selected problem includes:

- clear problem restatements,
- key invariants / observations,
- pseudocode,
- correctness sketches,
- complexity analysis,
- C++ (primary) and Python (secondary) implementations.

The focus is on clarity, correctness, and reasoning.

1.1 How to Use This Book

Problems include an optional **Author's Thinking** block, the authors plain language thinking and problem framing, which is excluded from exemplar builds.

Use this book to understand how to formally reason with and structure algorithm and data structures problems.

2 Arrays & Sequences

Arrays are the foundational data structure in most algorithmic problems. They provide constant-time indexed access and strong cache locality, but require careful reasoning about indices, bounds, and in-place mutation.

This chapter focuses on recognizing array-based problem patterns and maintaining clear invariants over contiguous memory.

2.1 Core Ideas

- Index-as-state reasoning
- In-place mutation vs auxiliary storage
- Prefix and suffix accumulation
- Simulation of elementary operations (e.g. addition, carry propagation)

2.2 Common Pitfalls

- Off-by-one errors at boundaries
- Unintended data overwrites during in-place updates
- Assuming sorting or uniqueness without verifying constraints
- Over-allocating space when a single pass suffices